

# Palmer Penguins Analysis

W7 skillset report

Jay Ballesteros

## Introduction

The Palmer Penguins dataset was collected from three islands near Palmer Station, Antarctica:

- Biscoe
- Dream
- Torgersen

The data originates from [this article](#).

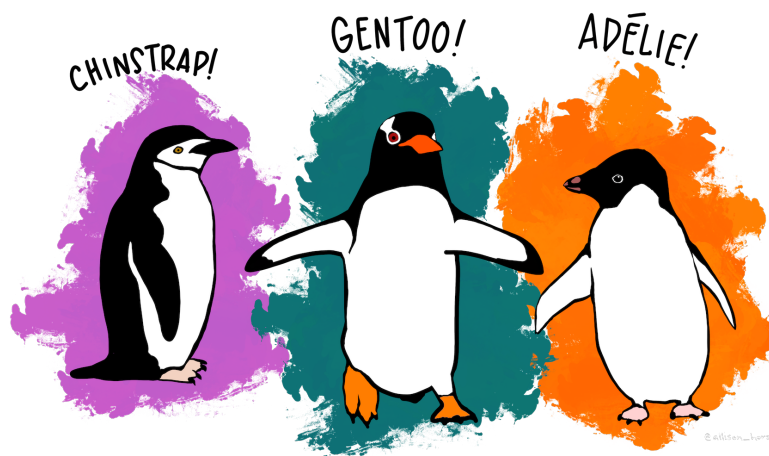


Figure 1: Penguin Species

## Data Analysis

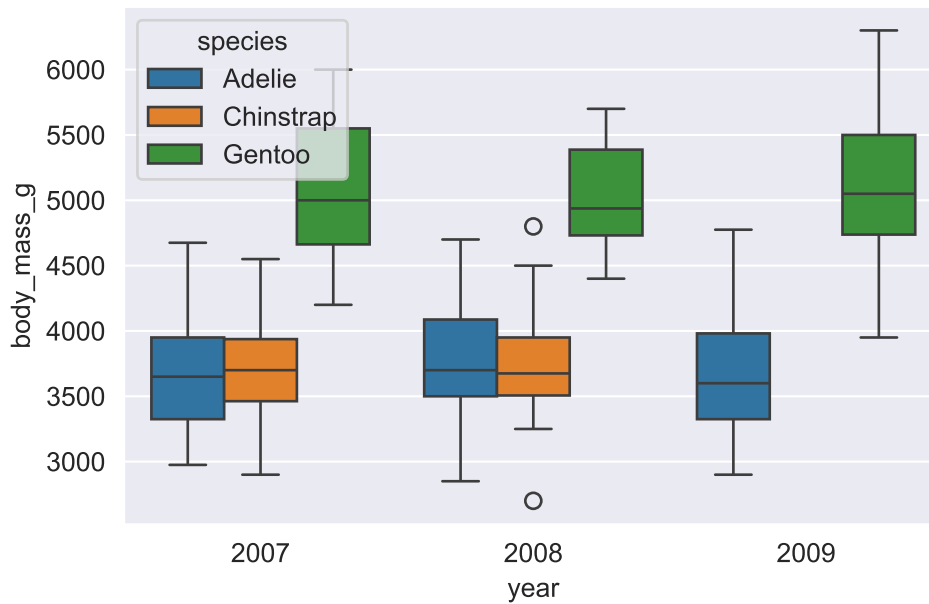
```
print(penguins.groupby("year")[["body_mass_g",  
"flipper_length_mm",  
"mean_temperature"]].mean().round(2))
```

|      | body_mass_g | flipper_length_mm | mean_temperature |
|------|-------------|-------------------|------------------|
| year |             |                   |                  |
| 2007 | 4031.65     | 196.42            | -1.61            |
| 2008 | 4085.75     | 200.10            | -0.78            |
| 2009 | 4468.70     | 205.82            | -1.76            |

```
print(penguins.groupby(["year", "species"])[["body_mass_g",  
"flipper_length_mm",  
"mean_temperature"]].mean().round(2))
```

|      |           | body_mass_g | flipper_length_mm | mean_temperature |
|------|-----------|-------------|-------------------|------------------|
| year | species   |             |                   |                  |
| 2007 | Adelie    | 3696.43     | 186.59            | -1.61            |
|      | Chinstrap | 3719.12     | 193.44            | -1.61            |
|      | Gentoo    | 5072.12     | 218.85            | -1.61            |
| 2008 | Adelie    | 3742.00     | 191.04            | -0.78            |
|      | Chinstrap | 3747.06     | 198.21            | -0.78            |
|      | Gentoo    | 5042.50     | 217.33            | -0.78            |
| 2009 | Adelie    | 3664.90     | 192.08            | -1.76            |
|      | Gentoo    | 5092.54     | 216.48            | -1.76            |

```
sns.boxplot(data=penguins, x="year", y='body_mass_g', hue='species')
```



Across all three years, Gentoo penguins are consistently the heaviest species. Interestingly, 2009 has the highest average body mass and flipper length despite having the coldest temperature ( $-1.76^{\circ}\text{C}$ ), while 2008 was the warmest year ( $-0.78^{\circ}\text{C}$ ) but had lower averages. This suggests that annual mean temperature alone does not explain changes in penguin body size, and other factors likely play a wider role.